

Benchmarks Are Engines, Not Cameras

How ML evaluation tools reshape the knowledge they measure

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Part 1: The engine metaphor

1. The Black-Scholes story
2. Camera vs. engine
3. Performativity and counterperformativity
4. Benchmarks do the same thing
5. The benchmark lifecycle

Part 2: Proof cultures

6. What is mechanistic interpretability?
7. Three communities, three standards
8. The denominator problem
9. Multiple comparisons and FWER

Part 3: Peer review makes it worse

10. The counterperformative loop
11. 11,196 ICLR submissions
12. Simpson's paradox explained
13. Simpson's paradox in our data

Part 4: What to do

14. Structural interventions
15. Project limitations
16. Conclusion

Part 1

Benchmarks are engines, not cameras

The Black-Scholes story

In 1973, Fischer Black and Myron Scholes published a formula for pricing options — the [Black-Scholes model](#).

Before it, options pricing was mostly intuition and guesswork. Different traders quoted wildly different prices for the same contract.

The formula said: given a stock price, volatility, time to expiry, and interest rate, there is **one correct price** for every option.

$$C = S \Phi(d_1) - K e^{-rT} \Phi(d_2)$$

The same year, the [Chicago Board Options Exchange](#) opened. Traders carried printouts of Black-Scholes pricing tables onto the trading floor.

The model made itself true

At first, actual option prices **deviated** from Black-Scholes predictions.

Then something strange happened. Over the next decade, market prices **converged** toward the formula's predictions.

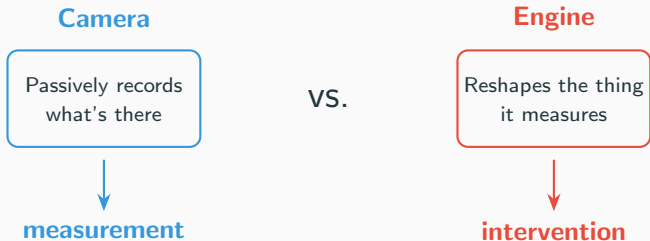


The formula didn't *describe* a pre-existing reality. Traders using the formula **created the reality it described**. And when the assumptions broke in 1987, prices diverged just as fast.

Camera vs. engine

The sociologist Donald MacKenzie studied this episode in *An Engine, Not a Camera* (2006) and asked: if a model **changes** the thing it measures, is it still a **measurement**?

The answer: no. The model is not a **camera** that passively records what's there. It's an **engine** that reshapes the world.



Our claim: benchmarks do the same thing

We argue that **ML benchmarks** operate exactly like Black-Scholes:

1. A benchmark is introduced to **measure** model capability.
2. Researchers optimize against it. Labs orient entire research programs around it.
3. The benchmark stops measuring a pre-existing ability — it starts **shaping** what gets built.
4. Eventually the benchmark saturates, and the community recognizes it was measuring the wrong thing all along.

The benchmark is not a camera. It's an engine.

We will now explore what that means — for statistical validity, for proof standards in ML, and for peer review.

Three levels of performativity

MacKenzie distinguishes how *deeply* a model can reshape the world:

Generic: Practitioners use the model in their work

Effective: Using the model *changes* the phenomenon it describes

Barnesian: The world conforms to the model — the map redraws the territory

stronger



Plus a failure mode: **counterperformativity** — when widespread adoption of the model causes the world to *diverge* from its predictions.

Concrete examples of each level

Level	Finance	ML
Generic	Traders use Black-Scholes to price options	Researchers use ImageNet to evaluate models
Effective	Using the formula changes how options are priced	Optimising for ImageNet changes what architectures get built
Barnesian	Market prices converge to the formula's predictions	"State of the art" = "best ImageNet score"
Counter-	Gaussian copula creates the correlated risk it assumed away	ML evaluation inflates the false positives it aims to prevent

The same instrument can be at different levels at different times. Black-Scholes was Barnesian in the 1980s and counterperformative in 1987.

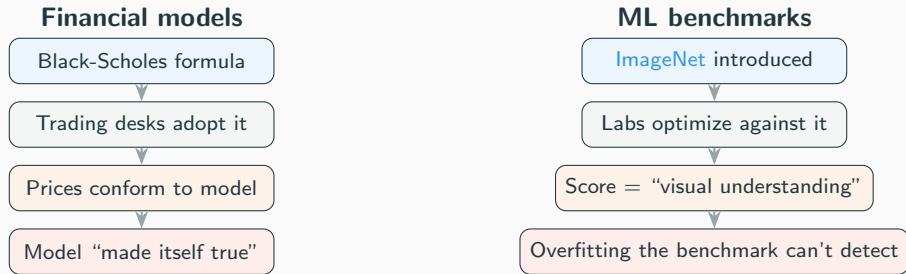
Counterperformativity: the Gaussian copula

The clearest example of a model **destroying** what it assumed:

1. David Li (2000) publishes the **Gaussian copula** formula for pricing mortgage-backed securities. Core assumption: defaults are only weakly correlated.
2. Every major bank adopts it. Trillions of dollars of CDOs are priced using this one model.
3. Because everyone uses the same model with the same assumption, they all hold the same risks. The **actual** correlation between defaults increases.
4. In 2007–08, correlated defaults cascade through the system. The model's assumption of low correlation **created** the high correlation that broke it.

The safety model generated the danger it was supposed to prevent. We argue the same structure operates in ML benchmarks.

Mapping this to ML benchmarks



The channels are different (trading desks vs. conference leaderboards), but the dynamic is the same: the instrument reshapes what it measures.

Where the analogy breaks (and where it holds)

Where it holds

- The instrument reshapes the thing it claims to measure
- Adoption is driven by institutional incentives, not just technical merit
- The community treats the model's output as ground truth
- Failure modes emerge from collective adoption, not individual misuse

Where it differs

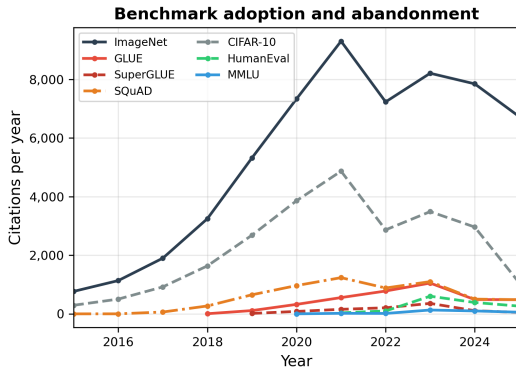
- No single formula — many benchmarks, loosely coordinated
- No direct financial feedback loop (prices don't clear)
- We claim **effective** performativity, not Barnesian — benchmarks reshape research priorities, not reality itself
- The “crash” is slow replacement, not a sudden crisis

We're not claiming $ML = \text{finance}$. We're claiming the **structure** is the same: an instrument presented as a camera that functions as an engine.

The benchmark lifecycle

Every benchmark goes through four phases:

1. **Camera.** Introduced as a measurement. Passively records model performance.
2. **Adoption.** Researchers optimize against it. Leaderboards emerge. Score = the capability.
3. **Saturation.** Optimisation changes what models are built. The benchmark reshapes training, data, architecture.
4. **Crash.** Benchmark is saturated or recognized as measuring the wrong thing. Replaced. Cycle restarts.



NLP benchmarks (GLUE, SuperGLUE) crash in 3–5 years.

Vision benchmarks (ImageNet) persist as cultural institutions.

NLP benchmarks crash fast

Benchmark	Domain	Introduced	Peak cites/yr	Decline to 2024
GLUE	NLP	2018	1,052	−54%
SuperGLUE	NLP	2019	359	−69%
SQuAD	NLP	2016	1,237	−59%
ImageNet	Vision	2009	9,305	−16%
CIFAR-10	Vision	2009	4,870	−39%

Key pattern: NLP benchmarks are abandoned when models saturate them — not when the underlying capability is “solved.”

GLUE wasn't replaced because language understanding was solved. It was replaced because BERT-scale models maxed out the score and the community needed something harder.

The benchmark tracks **social utility** — the ability to rank models.

Part 2

Proof cultures and the denominator problem

What is mechanistic interpretability?

The goal: understand **how** neural networks compute, not just *what* they compute.

Standard ML evaluation

- Run model on test set
- Report accuracy, F1, BLEU, etc.
- The model is a **black box**

This tells you *what* the model does, but not *why* or *how*.

That disagreement is the subject of Part 2.

Mechanistic interpretability

- Open up the model
- Identify internal components (features, circuits, causal variables)
- Explain **how it works**

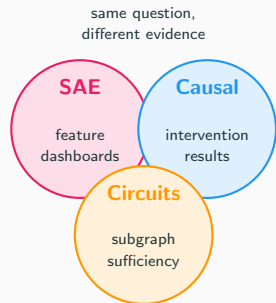
This is a young field. There is not yet broad agreement on what counts as evidence.

What is a “proof culture”?

MacKenzie studied [formal verification](#): what counts as a valid “proof” that a computer system is correct.

He found that **logicians**, **mathematicians**, and **engineers** accepted different kinds of arguments as proof. The boundary was shaped by funding and institutional authority, not just by logic.

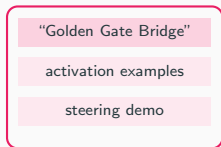
The same thing is happening in mechanistic interpretability. Three communities are trying to answer the same question — “how do neural networks compute?” — using **incompatible evidence standards**.



Three communities, three evidence standards

Sparse Autoencoders

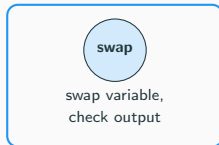
Feature dashboard



Valid if features are **interpretable**
and monosemantic.

Causal Abstraction

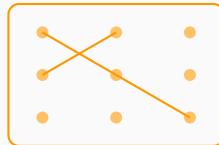
Intervention result



Valid if **interventions** produce
predicted effects.

Circuit Discovery

Circuit subgraph



Valid if the circuit is **necessary and sufficient** for the behavior.

These are not interchangeable. A feature dashboard would not satisfy a causal abstraction reviewer; an intervention result would not satisfy a circuits reviewer. Each community has its own "what counts as evidence."

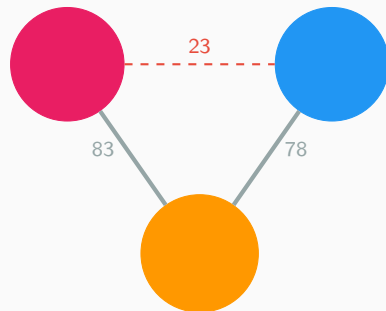
Subfields don't cite each other

We analyzed citation links among 688 ML papers (209 SAE, 40 Causal, 128 Circuits, 311 general).

From \ To	SAE	Causal	Circuits
SAE	326	14	45
Causal	9	63	16
Circuits	38	62	157

SAE $\leftrightarrow$ Causal: **23 cross-citations** across 16,720 possible directed pairs.

That's **0.14% density**. Both communities cite circuits literature, but barely cite each other.



Same foundation, no dialogue

The denominator problem

An SAE trains a dictionary of N features. The researcher analyzes them and ultimately chooses to display the k most interesting ones.

This is a multiple hypothesis test with $m \geq N$.

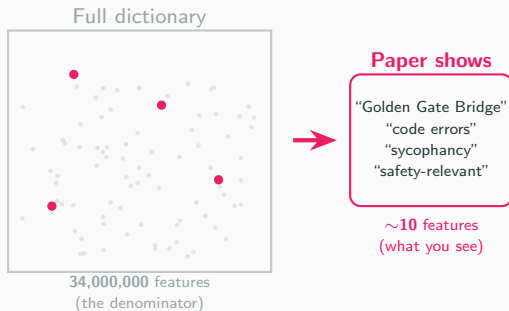
None of the papers we audited corrects for it.

The worked example:

[Templeton et al. \(2024\)](#) trained **34 million** features on Claude 3 Sonnet. They showed ~ 10 in detail.

Selection ratio: **3.4 million to one**.

If even 0.01% of features are spuriously interpretable, that's **3,400 false positives** — dwarfing the 10 shown.



The practitioner's funnel

Even outside ML, the denominator is invisible. A typical ML project:



The paper reports 15 comparisons. The reviewer sees 15. The reader sees 15. But 200+ were run. **The denominator is invisible by default.**

The multiple comparisons problem

A simple thought experiment:

Flip a fair coin 20 times. The chance of getting heads at least once is not 5% — it's **64%**.

Now imagine running 20 experiments, each with a 5% false positive rate. The chance that *at least one* gives a false positive is also 64%.



This is well understood in medicine and genetics. Researchers routinely correct for it. In ML, almost nobody does.

Metric 1: Family-wise error rate (FWER)

The probability that *at least one* of your comparisons is a false positive.

$$\text{FWER} = 1 - (1 - \alpha)^m$$

At $\alpha = 0.05$:

Comparisons (m)	FWER	
5	23%	(typical single experiment)
20	64%	(typical paper)
100	99.4%	(hyperparameter search)
1,000	$\approx 100\%$	(architecture search)

With enough comparisons, finding “significant” results is guaranteed by chance alone. Controlling FWER means ensuring this probability stays below α .

Metric 2: False discovery rate (FDR)

The expected *proportion* of false positives among your claimed discoveries.

$$\text{FDR} = \mathbb{E} \left[\frac{\text{false positives}}{\text{total rejections}} \right]$$

At $q = 0.05$, no matter how many tests you run:

Tests	Discoveries	Expected false	
20	4	~0.2	(5% of 4)
100	20	~1	(5% of 20)
1,000	50	~2.5	(5% of 50)
10,000	200	~10	(5% of 200)

FDR scales with discoveries, not total tests. It's less strict than FWER — you accept some false positives — but you find far more real effects. The tradeoff: you know ~5% of your results are wrong, but not *which* 5%.

Control 1: Bonferroni correction (1936)

The simplest fix has existed since 1936:

If you run m comparisons, divide your significance threshold by m .

$$\alpha_{\text{corrected}} = \frac{\alpha}{m} = \frac{0.05}{m}$$

Field	Typical m	Corrected α	Correction used?
Clinical trials	3–5	0.01	Always (FDA requires it)
Genomics (GWAS)	1,000,000	5×10^{-8}	Always
Psychology	5–20	0.003–0.01	Usually
ML (standard)	100–1,000+	would be 10^{-4}	Almost never
ML (MI/SAEs)	1,000–34,000,000	would be 10^{-9}	Never

Control 2: Holm–Bonferroni method (1979)

A strictly more powerful version of Bonferroni:

1. Sort your m p-values from smallest to largest: $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$
2. Compare each $p_{(i)}$ against a *different* threshold: $\frac{\alpha}{m-i+1}$
3. Reject from the smallest up, stop at the first non-rejection

Rank	P-value	Bonferroni threshold	Holm threshold
1st	0.003	$0.05/5 = 0.010$	$0.05/5 = 0.010$
2nd	0.009	$0.05/5 = 0.010$	$0.05/4 = 0.0125$
3rd	0.02	$0.05/5 = 0.010$	$0.05/3 = 0.017$

Holm rejects *at least* as many hypotheses as Bonferroni. It is uniformly more powerful while still controlling FWER. There is no reason to use Bonferroni over Holm.

Control 3: Benjamini–Hochberg procedure (1995)

A different philosophy: control the **false discovery rate** (FDR) instead of FWER.

FWER: probability of *any* false positive $\leq \alpha$

FDR: expected *proportion* of false positives among rejections $\leq q$

1. Sort p-values: $p_{(1)} \leq \dots \leq p_{(m)}$
2. Find the largest k such that $p_{(k)} \leq \frac{k}{m} \cdot q$
3. Reject all hypotheses $1, \dots, k$

This is much less conservative than Bonferroni/Holm. If you run 1,000 tests at $q = 0.05$, you accept that ~ 50 of your “discoveries” may be false — but you find far more real effects.

Widely used in genomics (thousands of genes tested simultaneously). Almost never used in ML.

Which correction to use?

Method	Controls	Best for
Bonferroni	FWER (strictest)	Very few tests; need zero false positives (clinical trials)
Holm	FWER (uniformly better)	Same situations as Bonferroni — strictly dominates it
Benjamini–Hochberg	FDR (more lenient)	Many tests; some false positives acceptable (screening, genomics)

For ML experiments (5–20 reported comparisons): Holm is the obvious default. It costs nothing, requires no assumptions beyond independence, and is trivial to implement.

For MI/SAE feature selection (thousands–millions of candidates): BH-style FDR control is more appropriate, since FWER becomes impossibly strict at that scale.

What other fields require

Field	Requirement	Enforced by
Clinical trials	Pre-registered correction (usually Bonferroni or Hochberg)	FDA / EMA (law)
Genomics (GWAS)	Genome-wide threshold 5×10^{-8} or BH-FDR	Journal policy
Psychology	Correction mandatory since replication crisis	APA guidelines
Neuroscience	Cluster-level FWE or FDR for imaging	Field norm

These aren't suggestions — in clinical trials, submitting results without pre-specified multiplicity adjustment is grounds for regulatory rejection. The FDA has mandated this since 1998.

ML has no equivalent authority, no pre-registration norm, and no review checklist that asks “how many comparisons did you actually run?”

What counts as a “comparison” in ML?

This is the key question. In ML, comparisons are often **implicit**:

- Every hyperparameter setting you tried
- Every architecture variant you tested
- Every random seed you ran (and kept the best)
- Every baseline you compared against
- Every dataset split you evaluated on
- Every feature you looked at and decided to show

A paper reporting “5 experiments” may have actually run 200+. The **denominator** — the total number of comparisons — is invisible to the reader.

This is not fraud. It’s the default workflow. But it has statistical consequences.

The scale of the problem in ML

Implicit comparison counts from real ML papers:

Paper	Comparisons (m)	FWER	Corrected?
Typical ML paper	100–500	>99%	No
IOI circuit (Wang+ 2022)	~1,000	$\approx 100\%$	No
SAEBench (Karvonen+ 2025)	~10,000	$\approx 100\%$	No
Scaling Monosemanticity (2024)	34,000,000	$\approx 100\%$	No

Every one of these papers reports “interesting” features or circuits. At these scales, finding interesting-looking results is *guaranteed* — even in random noise.

The question is not whether SAE features look interpretable. It’s whether the false positives are a problem.

How genomics solved this

Genome-wide association studies (GWAS) test $\sim 1,000,000$ genetic variants per study. Same scale as MI. Their solution (standardized since ~ 2005):

1. **Pre-register** the number of tests before looking at data
2. Apply BH-FDR at $q = 0.05$ for discovery (screening phase)
3. Replicate top hits in an **independent cohort** (validation phase)
4. Only replication survivors get published as “findings”

Result: genomics went from a replication crisis in the early 2000s to one of the most reliable fields in science. The false positive rate dropped from $>50\%$ to $<5\%$ within a decade.

MI could adopt the same pipeline: screen features/circuits with FDR control, then validate top candidates on held-out models or tasks. The infrastructure exists. The norm doesn't.

What ML papers could actually do

Realistic corrections that cost almost nothing to implement:

1. **Report the denominator.** “We trained an SAE with 65k features. We inspected 200. We show 10.” Just stating this changes the reader’s interpretation.
2. **Apply BH-FDR to any quantitative screen.** If you rank features by a metric (e.g. activation sparsity, probe accuracy), apply BH at $q = 0.05$. One line of code:
`statsmodels.stats.multitest.multipletests(pvals, method='fdr_bh')`
3. **Pre-register your hypothesis.** Decide what you’re looking for *before* looking at the features. Even informally (“we predict layer 8 contains a gender direction”) separates confirmation from exploration.

None of these require changing the research. They require changing the reporting.

Part 3

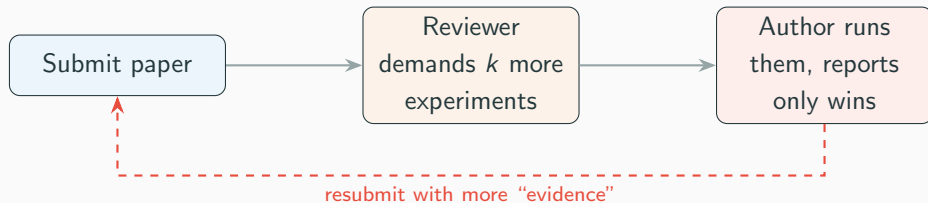
The counterperformativity of peer review

The paradox

Reviewers demand more experiments to **increase confidence**.

Each demanded experiment is a new comparison whose result is **selectively reported**.

The mechanism designed to reduce false positives may **structurally increase** them.



This is **counterperformativity**: the safety mechanism generates the risk it is designed to prevent. Same structure as the Gaussian copula in economics.

The data: 11,196 ICLR submissions

We analyzed every public submission to [ICLR 2024](#) ($n = 7,404$) and [ICLR 2023](#) ($n = 3,792$).

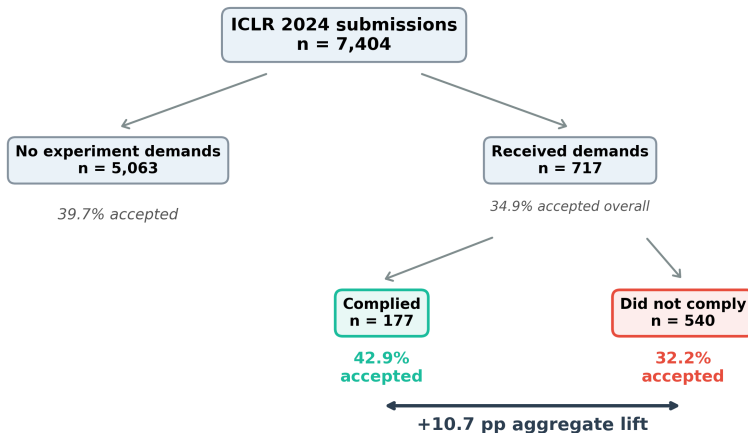
For each paper, we extracted:

- Experiment demands from reviewer text
- Compliance signals from author rebuttals
- Final accept/reject decision

Headline: papers that comply with experiment demands are accepted at **+10.7 pp** higher rates than those that don't.

Sounds like the system works, right?

Demand compliance and acceptance



Simpson's paradox (score-stratified, n = 5,780):
Low (<4.5): 0.0% vs 0.0% = 0.0 pp
Borderline: 7.5% vs 4.7% = +2.9 pp
Mid (5.5-6.5): 50.0% vs 54.7% = -4.7 pp
High (>6.5): 94.9% vs 94.8% = +0.1 pp
Aggregate lift is a composition effect.

What is Simpson's paradox?

A trend that appears in aggregate data **reverses** when you split the data into subgroups.

Classic example: Berkeley admissions

In 1973, UC Berkeley was sued for gender bias: overall, men were admitted at higher rates than women.

But when you looked *department by department*, women were admitted at **equal or higher** rates in most departments.

The aggregate gap existed because women applied to more competitive departments.

The lesson

An aggregate statistic can tell a completely different story from the stratified data.

This matters whenever subgroups have different sizes *and* different base rates.

We found exactly this pattern in ICLR compliance data — and it changes the interpretation entirely.

Simpson's paradox: the lift is a composition effect

The +10.7 pp aggregate lift is a **Simpson's paradox**. When you stratify by reviewer score:

Stratum	Comply	No comply	Lift
Low (<4.5)	0.0%	0.0%	0 pp
Border (4.5–5.5)	7.5%	4.7%	+2.9 pp
Mid (5.5–6.5)	50.0%	54.7%	−4.7 pp
High (≥6.5)	94.9%	94.8%	+0.1 pp
Aggregate	42.9%	32.2%	+10.7 pp

The lift **vanishes** within strata. The aggregate effect comes from composition: compliant papers cluster in the borderline stratum.

Why this is worse, not better:

The compliance incentive concentrates **exactly where selective reporting is most consequential** — borderline papers, where one more supportive experiment can tip the decision.

For low-scored papers: compliance is **futile**.

For high-scored papers: compliance is **unnecessary**.

For borderline papers: compliance is **most tempting** and **least detectable**.

Why this matters: performativity driven by a mirage

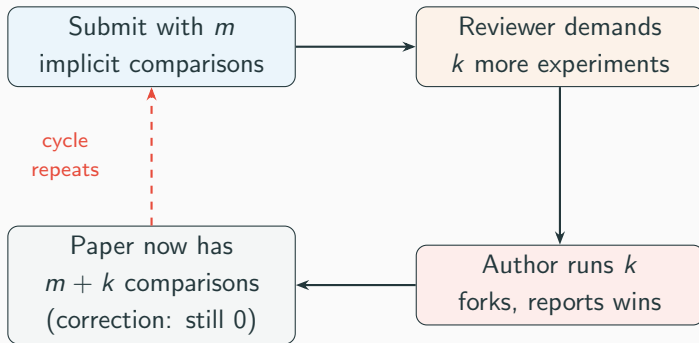
The aggregate data tells researchers: “comply with experiment demands → get accepted.” But the stratified data shows this is an illusion — compliance doesn’t actually help within any score band. So what’s happening?

1. Researchers *believe* more experiments = acceptance (the aggregate signal)
2. They optimize for compliance: run more benchmarks, add more ablations
3. This doesn’t actually improve their acceptance odds (the stratified signal)
4. But it *does* inflate the total number of implicit comparisons
5. Which inflates the FWER — more false positives enter the literature

The review system reshapes research behavior based on a statistical artifact. Researchers game a signal that isn’t real — and the gaming itself causes harm (more uncorrected comparisons).

This is the counterperformative loop:

The counterperformative loop



Each cycle adds comparisons
without adjusting the denominator.
The FWER grows invisibly.

Part 4

What to do about it

Structural interventions

Make denominators visible

- Report total experiments run, not just those in the paper
- For SAEs: report interpretability *rates*, not cherry-picked features
- Template: “trained N features, k scored above threshold t ”

Audit demands for FWER

- Before demanding an ablation, ask whether it increases the comparison count
- Demand the correction alongside the experiment

Publish negative results

- GDM’s “research we didn’t publish” post is the template
- Venues should create tracks for replications and null results

Bridge proof cultures

- Cross-community benchmarks ([SAEBench](#), [MIB](#) are a start)
- Shared evaluation criteria across subfields
- Require engagement with alternative evidence standards in reviews

Project limitations

1. **Observational.** The demand-compliance analysis is correlational. Simpson's paradox controls for quality, but within-stratum confounds remain.
2. **One venue.** ICLR is the only major venue that publishes rejected submissions. Can't replicate at NeurIPS or ICML without the full corpus.
3. **Effective, not Barnesian.** We claim benchmarks *reshape* research priorities, not that the world conforms to the benchmark in the strong Black-Scholes sense.
4. **No ethnography.** A proper sociology of ML would require interviews and lab observation, not just citation analysis.
5. **Infrastructure gap.** Even if norms changed, experiment-tracking tools default to private. The tooling for public denominator reporting doesn't exist yet.

Statistical reform treats the **symptom**.

The cause is an evaluation culture whose
instruments both **measure** and
manufacture what counts as knowledge.

The benchmark is an engine, not a camera.

And like all engines, it needs maintenance, inspection,
and occasionally — replacement.

Thank you

Paper: TBD

Repository: github.com/elliottower/benchmarks-engines

Contact: elliott@elliottower.ai

Key references:

- MacKenzie (2006). *An Engine, Not a Camera*. MIT Press.
- MacKenzie (2001). *Mechanizing Proof*. MIT Press.
- MacKenzie & Bamford (2018). *Counterperformativity*. *New Political Economy*.
- Espeland & Sauder (2007). *Rankings and reactivity*. *American Journal of Sociology*.

Data: 11,196 ICLR submissions, 18 manual paper audits, OpenAlex citation analysis, Semantic Scholar citation network of 27 ML papers.